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1. Quantifying Information Stored in Synaptic Connections Rather Than in Firing Activities of Neural Networks

Xinhao Fan, Shreesh P. Mysore

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A cornerstone of our understanding of both biological and artificial neural networks is that they store information in the strengths of synaptic connections among the neurons. However, in contrast to the well-established theory for quantifying information encoded by the firing activity of neural networks, there does not exist a framework for quantifying information stored in the network's connection distribution itself. Here, we develop a theoretical framework for synaptic information by using densely connected Hebbian networks performing autoassociative memory tasks and by modeling data patterns to be stored as log-normal distributions. Specifically, we derive analytical approximations for Shannon mutual information between the data and singletons, pairs, and arbitrary n-tuples of synaptic connections within the network. Our framework corroborates well-established insights regarding pattern storage capacity, supports the principle of distributed coding in neural firing activities, and formalizes the heterogeneity inherent in information encoding across synapses in a network. Notably, it discovers synergistic interactions among synapses, revealing that the information encoded jointly by all the synapses exceeds the sum of its parts. Taken together, this study introduces a powerful, interpretable framework for quantitatively understanding information storage in the synapses of neural networks, one that illustrates the duality of synaptic connectivity and neural population activity in learning and memory.

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2. Associative Memory and Generative Diffusion in the Zero-Noise Limit

Joshua Hess, Quaid Morris

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This letter shows that generative diffusion processes converge to associative memory systems at vanishing noise levels and characterizes the stability, robustness, memorization, and generation dynamics of both model classes. Morse-Smale dynamical systems are shown to be universal approximators of associative memory models, with diffusion processes as their white-noise perturbations. The universal properties of associative memory that follow are used to characterize a generic transition from generation to memory as noise diminishes. The structural stability of Morse-Smale flows-that is, the robustness of their global critical point structure-implies the stability of both trajectories and invariant measures for diffusions in the zero-noise limit. The learning and generation landscapes of these models appear as parameterized families of gradient flows and their stochastic perturbations, and the bifurcation theory for Morse-Smale systems implies that they are generically stable except at isolated parameter values, where enumerable sets of local and global bifurcations govern transitions between stable systems. These landscapes are thus characterized by ordered bifurcation sequences that create, destroy, or alter connections between rest points and are robust under small stochastic or deterministic perturbations. The framework is agnostic to model formulation, which we verify with examples from energy-based models, denoising diffusion models, and classical and modern Hopfield networks. We additionally derive structural stability criteria for Hopfield-type networks and find that

simple cases violate them. Collectively, our geometric approach provides insight into the classification, stability, and emergence of memory and generative landscapes.

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3. Hierarchical Active Inference Using Successor Representations

Prashant Rangarajan, Rajesh P. N. Rao

https://doi.org/10.1162/neco.a.1545 | pp. 1-54

Active inference, a neutrally inspired model for inferring actions based on the free energy principle (FEP), has been proposed as a unifying framework for understanding perception, action, and learning in the brain. Active inference has previously been used to model ecologically important tasks such as navigation and planning, but scaling it to solve complex large-scale problems in real-world environments has remained a challenge. Inspired by the existence of multiscale hierarchical representations in the brain, we propose a model for planning actions based on hierarchical active inference. Our approach combines a hierarchical model of the environment with successor representations for efficient planning. We present results demonstrating (1) how lower-level successor representations can be used to learn higher-level abstract states, (2) how planning based on active inference at the lower level can be used to bootstrap and learn higher-level abstract actions, and (3) how these learned higher-level abstract states and actions can facilitate efficient planning. We illustrate the performance of the approach on several planning and reinforcement learning (RL) problems, including a variant of the well-known four rooms task, a key-based navigation task, a partially observable planning problem, the mountain car problem, and PointMaze, a family of navigation tasks with continuous state and action spaces. Our results represent, to our knowledge, the first application of learned hierarchical state and action abstractions to active inference in FEP-based theories of brain function.

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4. Shape Matters: Few-Shot Object Classification From High-Information Contour Features

Maria Osório, Alexandre Bernardino, Andreas Wichert

https://doi.org/10.1162/neco.a.1563 | pp. 1-22

A key challenge in visual object recognition is developing models that generalize from limited data while maintaining transparency in their decision making. We propose a biologically inspired model that addresses both issues by classifying images based on transformation-invariant local shape key features. Following the principles of the brain's what and where pathways, each feature is encoded by an image patch and its relative location in polar coordinates, enabling interpretable and robust comparisons between inputs and class prototypes. To mimic human concept learning, prototypes are selected using clustering, improving representativeness and generalization. Results show that our model achieves human-comparable performance, with an error rate between 1% and 2% on the MNIST data set when all training images are used as prototypes. In data-limited scenarios, where only a small number of prototypes are selected, our model consistently outperforms convolutional neural networks (CNNs). To evaluate out-of-distribution generalization, we use prototypes from MNIST and test both models on the ETL-1 data set, which differs in data distribution. Although CNN accuracy drops significantly under these conditions, our model maintains high accuracy, even with few prototypes, demonstrating strong robustness and greater capacity to generalize to unseen distributions, bringing it closer to human-like recognition capabilities.

5. Neuronal Spike Trains as Functional-Analytic Distributions: Representation, Analysis, and Significance

Gabriel A. Silva

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The action potential constitutes the digital component of the signaling dynamics of neurons. But the biophysical nature of the full-time course of the action potential associated with changes in membrane potential is mathematically distinct from its representation as a discrete set of events that encode when action potentials are triggered in a collection of spike trains. In this letter, we develop from first principles a unified functional-analytic framework for neuronal spike trains, grounded in Schwartz distribution theory. We show how this representation provides an exact operational calculus for convolution, distributional differentiation, and distributional support, which enables closed-form analysis of spike train dynamics without discretization, rate approximation, or smoothing. We then analyze the framework in the context of a two-neuron reciprocal circuit with propagation latencies and refractoriness, deriving exact results for synaptic drive, spike timing sensitivity, and causal admissibility of inputs, quantities that are either ill-defined or require approximation in conventional treatments.

6. Unsupervised Feature Selection Using Bayesian Tucker Decomposition

Y-h. Taguchi, Yoh-ichi Mototake

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In this letter, we propose Bayesian Tucker decomposition (BTuD) in which the residuals are supposed to follow. Although we have proposed an algorithm to perform the proposed BTuD, the conventional higher-order orthogonal iteration can generate Tucker decomposition consistent with the present implementation. Using the proposed BTuD, we can perform unsupervised feature selection successfully applied to various synthetic data sets, global coupled maps with randomized coupling strength, and gene expression profiles. Thus, we can conclude that our newly proposed unsupervised feature selection method is promising. In addition to this, BTuD-based unsupervised feature extraction (FE) is expected to coincide with TD-based unsupervised FE that were previously proposed and successfully applied to a wide range of problems.

7. Attractor-Based Models for Sequences and Pattern Generation in Neural Circuits

Juliana Londono Alvarez, Katherine Morrison, Carina Curto

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Neural circuits in the brain perform a variety of essential functions, including input classification, pattern completion, and the generation of rhythms and oscillations that support functions such as breathing and locomotion. There is also substantial evidence that the brain encodes memories and processes information via sequences of neural activity. Traditionally, rhythmic activity and pattern generation have been modeled using coupled oscillators, whereas input classification and pattern completion have been modeled using attractor neural networks. Here, we present a theoretical framework that demonstrates how attractor-based

networks can also generate diverse rhythmic patterns, such as those of central pattern generator circuits (CPGs). Additionally, we propose a mechanism for transitioning between patterns. Specifically, we construct a network that can step through a sequence of five different quadruped gaits. It is composed of two dynamically distinct modules: a "counter" network that can count the number of external inputs it receives via a sequence of fixed points and a locomotion network that encodes five different quadruped gaits as limit cycles. A sequence of locomotive gaits is obtained by connecting the counter network with the locomotion network. Specifically, we introduce a new architecture for layering networks that produces fusion attractors, binding pairs of attractors from individual layers. All of this is accomplished within a unified framework of attractor-based models using threshold-linear networks.

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8. Neuromodulators Generate Multiple Context-Relevant Behaviors in Recurrent Neural Networks

Ben Tsuda, Stefan Carl Pate, Kay M. Tye, Hava T. Siegelmann, Terrence J. Sejnowski
https://doi.org/10.1162/neco.a.1489 | pp. 292-327

Neuromodulators are critical controllers of neural states, with dysfunctions linked to various neuropsychiatric disorders. Although many biological aspects of neuromodulation have been studied, the computational principles underlying how neuromodulation of distributed neural populations controls brain states remain unclear. In contrast to external contextual inputs, neuromodulation can act as a single scalar signal that is broadcast to a vast population of neurons. We model the modulation of synaptic weight in a recurrent neural network model and show that neuromodulators can dramatically alter the function of a network, even when highly simplified. We find that under structural constraints like those in brains, this provides a fundamental mechanism that can increase the computational capability and flexibility of a neural network. Diffuse synaptic weight modulation enables storage of multiple memories using a common set of synapses that are able to generate diverse, even diametrically opposed, behaviors. Our findings help explain how neuromodulators unlock specific behaviors by creating task-specific hyperchannels in neural activity space and motivate more flexible, compact and capable machine learning architectures.

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9. Object Detection, Recognition, Deep Learning, and the Universal Law of Generalization

Faris Rustom, Rohan Sharma, Haluk Ö■men, Arash Yazdanbakhsh
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Object detection and recognition are fundamental functions that play a significant role in the success of species. Because the appearance of an object exhibits large variability, the brain has to group these different stimuli under the same object identity, a process of generalization. Does the process of generalization follow some general principles, or is it an ad hoc bag of tricks? The universal law of generalization (ULoG) provides evidence that generalization follows similar properties across a variety of species and tasks. Here, we tested the hypothesis derived from ULoG that the internal representations underlying generalization reflect the natural properties of object detection and recognition in our environment rather than the specifics of the system solving these problems. Neural networks with universal-approximation capability have been successful in many object detection and recognition tasks; however, how these networks reach their decisions remains opaque. To provide a strong test for ecological

validity, we used natural camouflage, which is nature's test bed for object detection and recognition. We trained a deep neural network with natural images of "clear" and "camouflaged" animals and examined the emerging internal representations. We extended ULoG to a realistic learning regime, with multiple consequential stimuli, and developed two methods to determine category prototypes. Our results show that with a proper choice of category prototypes, the generalization functions are monotone decreasing, similar to the generalization functions of biological systems. Critically, we show that camouflaged inputs are not represented randomly but rather systematically appear at the tail of the monotone decreasing functions. Our results support the hypothesis that the internal representations underlying generalization in object detection and recognition are shaped mainly by the properties of the ecological environment, even though different biological and artificial systems may generate these internal representations through drastically different learning and adaptation processes. Furthermore, the extended version of ULoG provides a tool to analyze how the system organizes its internal representations during learning as well as how it makes its decisions.

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10. Exploring the Interplay Between BOLD Signal Variability, Complexity, Static and Dynamic Functional Brain Network Features During Movie Viewing

Amirhossein Ghaderi, Hongye Wang, Andrea B. Protzner

https://doi.org/10.1162/neco.a.1488 | pp. 373-402

Exploring the dynamics and complexity of brain signal is critical to advancing our understanding of brain function. Recent fMRI studies have revealed links between BOLD signal variability or complexity with static/dynamics features of functional brain networks (FBN). However, the association between variability/complexity and regional centrality is still understudied. Here we investigate the association between variability/complexity and static/dynamic nodal features of FBN using graph theory analysis with fMRI BOLD data acquired during naturalistic movie watching. We found that variability positively correlated with fine-scale complexity but negatively correlated with coarse-scale complexity. Specifically, regions with high centrality and clustering coefficient were related to less variable but more complex signal. Similar relationships persisted for dynamic FBN, but the associations with certain aspects (e.g., eigenvector centrality) of regional centrality dynamics became insignificant. Our findings demonstrate that the relationship between BOLD signal variability and static/dynamic FBN with BOLD signal complexity depends on the temporal scale of signal complexity and that time-varying features of FBN reflect the complexities of how BOLD signal variability/complexity coevolve with dynamic FBN.

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11. Local Glutamate-Glutamine Cycling Underlies Presynaptic ATP Homeostasis

Reinoud Maex

https://doi.org/10.1162/neco.a.1490 | pp. 403-438

Presynaptic axon terminals maintain in their cytosol an almost constant level of adenosine triphosphate (ATP) to safeguard neurotransmission during varying workloads. In the study reported in this letter, it is argued that the vesicular release of neurotransmitter and the recycling of transmitter via astrocytes may itself be a mechanism of ATP homeostasis. In a minimal metabolic model of a presynaptic axon bouton, the accumulation of glutamate into vesicles and the activity-dependent supply of its precursor glutamine by astrocytes generated a steady-state level of ATP that was independent of the workload. When the workload

increased, an enhanced supply of glutamine raised the rate of ATP production through the conversion of glutamate to the Krebs cycle intermediate α-ketoglutarate. The accumulation and release of glutamate, on the other hand, acted as a leak that diminished ATP production when the workload decreased. The fraction of ATP that the axon spent on the release and recycling of glutamate was small (4.7%), irrespective of the workload. Increasing this fraction enhanced the speed of ATP homeostasis and reduced the futile production of ATP. The model can be extended to axons releasing other, or coreleasing multiple, transmitters. Hence, the activity-dependent formation and release of neurotransmitter may be a universal mechanism of ATP homeostasis.

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12. Reframing the Expected Free Energy: Four Formulations and a Unification

Théophile Champion, Howard Bowman, Dimitrije Markovi■, Marek Grzes

https://doi.org/10.1162/neco.a.1491 | pp. 439-469

Active inference is a process theory of perception, learning, and decision making that is applied to a range of research fields, including neuroscience, robotics, psychology, and machine learning. Active inference rests on an objective function called the expected free energy, which can be justified by the intuitive plausibility of its formulations-for example, the risk plus ambiguity and information gain/pragmatic value formulations. This letter seeks to formalize the problem of deriving these formulations from a single root expected free energy definition-the unification problem. Then we analyze two approaches to defining expected free energy. More precisely, the expected free energy is either defined as (1) the risk over observations plus ambiguity or (2) the risk over states plus ambiguity. In the first setting, no rigorous mathematical justification for the expected free energy has been proposed to date, but all the formulations can be recovered from it by assuming that the likelihood of target distribution T(o|s) is the likelihood of the generative model P(o|s). Importantly, under this likelihood constraint, if the likelihood is lossless,1 then prior preferences over observations can be defined arbitrarily. However, in the more general case of partially observable Markov decision processes (POMDPs), we demonstrate that the likelihood constraint effectively restricts the set of valid prior preferences over observations. Indeed, only a limited class of prior preferences over observations is compatible with the likelihood mapping of the generative model. In the second setting, a justification of the root expected free energy definition exists, but this setting only accounts for two formulations: the risk over states plus ambiguity and entropy plus expected energy formulations. We conclude with a discussion of the conditions under which a unification of expected free energy formulations has been proposed in the literature by appeal to the free energy principle in the specific context of systems without random fluctuations.

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13. Inhibitory Feedback Enables Predictive Learning of Multiple Sequences in Neural Networks

Matteo Saponati, Martin Vinck

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Anticipating future events is a key computational task for neuronal networks. Experimental evidence suggests that reliable temporal sequences in neural activity play a functional role in the association and anticipation of events in time. However, how neurons can differentiate and anticipate multiple spike

sequences remains largely unknown. We implement a learning rule based on predictive processing, where neurons exclusively fire for the initial, unpredictable inputs in a spiking sequence, leading to an efficient representation with reduced postsynaptic firing. Combining this mechanism with inhibitory feedback leads to sparse firing in the network, enabling neurons to selectively anticipate different sequences in the input. We demonstrate that intermediate levels of inhibition are optimal to decorrelate neuronal activity and to enable the prediction of future inputs. Notably, each sequence is independently encoded in the sparse, anticipatory firing of the network. Overall, our results demonstrate that the interplay of self-supervised predictive learning rules and inhibitory feedback enables fast and efficient classification of different input sequences.

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14. Potential for Reinforcement Learning in the Cerebellum

Richard W. Prager, Richard Apps

https://doi.org/10.1162/neco.a.1507 | pp. 499-537

This article explores how simple reinforcement learning algorithms might be implemented by the anatomy of the cerebellum. In doing this, we highlight which anatomical and physiological details are most important for assessing algorithmic fit, and we discuss which algorithm components are easiest to accommodate in a neural system. We describe hypothetical cerebellar implementations of four reinforcement learning algorithms and discuss the anatomical plausibility of the various components required. We show how one of the algorithms can learn to generate short sequences of actions without continuous information on the resulting changes to the environment. We finish with simulations that illustrate the way that the algorithms learn to solve the problem of balancing an inverted pendulum, commonly known as the cart-pole problem. We highlight two physiological features: reward signals and combining information across time, that indicate that some sort of reinforcement learning adaptation may be taking place. We also describe why the commonly used algorithmic feature, an eligibility trace, presents particular problems to implement in known neural anatomy.

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15. The Cooperative Network Architecture: Learning Structured Networks as Representation of Sensory Patterns

Pascal Sager, Jan M. Deriu, Benjamin F. Grewe, Thilo Stadelmann, C. von der Malsburg

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We introduce the cooperative network architecture (CNA), a model that represents sensory signals using structured, recurrently connected networks of neurons, termed "nets." Nets are dynamically assembled from overlapping net fragments, which are learned based on statistical regularities in sensory input. This architecture offers robustness to noise, deformation, and generalization to out-of-distribution data, addressing challenges in current vision systems from a novel perspective. We demonstrate that net fragments can be learned without supervision and flexibly recombined to encode novel patterns, enabling figure completion and resilience to noise. Our findings establish CNA as a promising paradigm for developing neural representations that integrate local feature processing with global structure formation, providing a foundation for future research on invariant object recognition.

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16. Force Learning in Balanced Cortical E-I Networks

Takashi Kanamaru, Kazuyuki Aihara

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Force learning is a learning method for generating various types of complex dynamics in recurrent neural networks (RNNs), which is related to the reservoir computing (RC). RC uses an RNN called reservoir whose synaptic weights are randomly generated and fixed during learning. Force learning trains these synaptic weights inside the reservoir networks. Although force learning can be used as an effective tool for machine learning, possibilities of its realization in the brain are not often discussed. Here, in order to consider the possibilities of its realization in the brain, force learning is applied to an excitatory and inhibitory (E-I) network that models the cerebral cortex. A multimodule network composed of excitatory and inhibitory neurons is defined, and a readout is put outside, similar to a conventional reservoir. The output of this network is calculated at the readout as a linear combination of the filtered average firing rates of the excitatory neurons in the modules. Feedback connections that provide output back to the excitatory neurons in the modules with random strength are also added to this network. This network typically shows transitive chaotic synchronization, in which synchronizing modules are rearranged chaotically and intermittently. Under such conditions, our E-I network is trained to generate sinusoidal periodic signals for simplicity with force learning. When adjusting the E-I activity, it is observed that the efficiency of force learning is maximized at an optimal E-I balance near an edge of chaos. These results imply that the cooperation of excitatory and inhibitory neurons is required when force learning works effectively in the brain, although usual reservoir networks don't distinguish these two kinds of neurons.

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17. Multiclass Linear Perceptrons With Multiplicative Margins

Dmitri A. Rachkovskij, Evgeny Osipov, Olexander Volkov, Daswin Pasantha De Silva, Denis Kleyko

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This article introduces a family of multiclass linear perceptron classifiers with a multiplicative margin mechanism (MMPerc), as an alternative to standard margin-free and additive margin perceptrons. The multiplicative formulation enforces classification confidence by requiring the true class score to exceed that of competing classes by a specified fraction of itself rather than by a fixed additive threshold. This avoids dependence on score magnitudes arising from varied norms of data and class weight vectors. We propose several architectural and algorithmic variants of MMPerc, derive associated loss functions and mistake bounds for both linearly separable and nonseparable data, and analyze key design considerations, including bias, margin threshold selection, and training modes. Extensive experiments on synthetic and real data sets show that MMPerc classifiers typically outperform the standard perceptron, as well as classic baselines such as support vector machines and ridge classifiers. Owing to their simplicity, minimalistic design, and computational efficiency, MMPerc classifiers are promising candidates for conventional machine learning tasks, linear evaluation of deep neural networks, integration with hyperdimensional computing and vector symbolic architecture representations, and deployment in resource-constrained applications.

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18. Cognitive Control Strategies Derive From Dimension Reliability

William H. Alexander

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To explain behavioral effects, models of cognitive control frequently rely on task information that the modeler provides. Hard-wired information can include labeling task dimensions as being relevant or irrelevant, defining which task stimuli belong to which task dimensions, or proposing a specific strategy by which control is adjusted during task performance. Although models incorporating hard-wired information of this nature are frequently successful at accounting for observed behavior, their ability to do so often depends on tailoring this information to specific tasks, usually performed in a laboratory setting. Outside of the laboratory, individuals are not usually provided explicit information about how to behave; it thus remains an open question as to how individuals identify, update, and switch task strategies in the real world. Here, we present a new model of cognitive control, learned attention for control (LAC), that not only captures a broad range of control effects but does so using a minimal amount of modeler-supplied information. In a series of simulations, we demonstrate how the LAC model adopts distinct control strategies based on recent trial history and adapts to changing behavioral contexts. The model's ability to do so derives from an ongoing evaluation of how well task stimuli independently predict correct behavior, and the results of this evaluation are used to shift attention among information sources. These results suggest that the reliability of information can serve as a general principle for understanding cognitive control.

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19. Similarity Matching Networks: Hebbian Learning and Convergence Over Multiple Timescales

Veronica Centorrino, Francesco Bullo, Giovanni Russo

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A recent breakthrough in biologically plausible normative frameworks for dimensionality reduction is based on the similarity matching cost function and the low-rank matrix approximation problem. Despite clear biological interpretation, successful application in several domains, and experimental validation, a formal complete convergence analysis remains elusive. Building on this framework, we consider and analyze a continuous-time neural network, the similarity matching network, for principal subspace projection. Derived from a min-max-min objective, this biologically plausible network consists of three coupled dynamics evolving at different timescales: neural dynamics, lateral synaptic dynamics, and feedforward synaptic dynamics at the fast, intermediate, and slow timescales, respectively. The feedforward and lateral synaptic dynamics consist of Hebbian and anti-Hebbian learning rules, respectively. By leveraging a multilevel optimization framework, we prove convergence of the dynamics in the offline setting. Specifically, at the first level (fast timescale), we show strong convexity of the cost function and global exponential convergence of the corresponding gradient-flow dynamics. At the second level (intermediate timescale), we prove strong concavity of the cost function and exponential convergence of the corresponding gradient-flow dynamics within the space of positive definite matrices. At the third and final level (slow timescale), we study a nonconvex and nonsmooth cost function, provide explicit expressions for its global minima, and prove almost sure convergence of the corresponding gradient-flow dynamics to the global minima. These results rely on two empirically motivated conjectures that are supported by thorough numerical experiments. Finally, we validate the effectiveness of our approach via a numerical example.

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20. Echoes of the Past: A Unified Perspective on Fading Memory and Echo States

Juan■Pablo Ortega, Florian Rossmannek
https://doi.org/10.1162/neco.a.1510 | pp. 765-782

Recurrent neural networks (RNNs) have become increasingly popular in information processing tasks involving time series and temporal data. A fundamental property of RNNs is their ability to create reliable input/output responses, often linked to how the network handles its memory of the information it processed. Various notions have been proposed to conceptualize the behavior of memory in RNNs, including steady states, echo states, state forgetting, input forgetting, and fading memory. Although these notions are often used interchangeably, their precise relationships remain unclear. This work aims to unify these notions in a common language, derive new implications and equivalences between them, and provide alternative proofs to some existing results. By clarifying the relationships between these concepts, this research contributes to a deeper understanding of RNNs and their temporal information processing capabilities.

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21. Signal-Dependent Planning Noise Reduces Task Interference by Assisting in the Formation of Stable Motor Primitives in a Neural Network Model of Muscle Coordination Learning

Daniel W. Feng, David J. Reinkensmeyer, Juan C. Pérez-Ibarra
https://doi.org/10.1162/neco.a.1512 | pp. 783-822

In human motor coordination, learning to coactivate multiple muscles at once to achieve distinct target combinations of forces or tasks remains a fundamental area of study. Task interference, where training on one task degrades performance on previously learned tasks, can slow motor learning. However, the neural mechanisms that reduce interference are not fully understood. We hypothesized that the structure of planning noise, specifically its signal-dependent nature, significantly shapes learning dynamics and limits interference within motor learning systems that rely on variability for exploration. To test this hypothesis, we developed a three-layer neural network model of muscle coordination informed by key neuroanatomical and neurophysiological principles and simulated learning for producing various combinations of muscle forces. Synaptic weights were stochastically altered from trial to trial with either fixed-variance planning noise (FVPN), where each connection's variance was fixed during learning, or signal-dependent planning noise (SDPN), where noise variance depended on the neuron population activity. Weights were reinforced when they reduced output error relative to target forces. An execution noise term, applied to the motor output, modeled peripheral motor variability. However, the learning rule was not informed about how much of the output corresponded to peripheral or central variability. Our results showed that SDPN improved both the rate and accuracy of multitask learning by reducing task interference compared to FVPN across network sizes, training schedules, and execution noise levels. SDPN achieved this by concentrating neural plasticity within the neuron populations engaged by the current task rather than modifying the entire network. This signal-dependent plasticity allowed multiple motor primitives to form, stabilize, and be reused for new tasks. The model replicated the well-documented benefit of interleaved versus blocked training in motor learning. As a computational proof of concept, this work suggests that SDPN can benefit multitask motor training by facilitating the formation of motor primitives.

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22. Infinite Horizon Control With Nonlinear Dynamics Models Reproduces Temporal Modulation of Reaching Movements

Antoine De Comite, Hari Teja Kalidindi, J. Andrew Pruszynski, Frédéric Crevecoeur
https://doi.org/10.1162/neco.a.1515 | pp. 823-844

Movement duration, a fundamental aspect of motor control, is often viewed as a preprogrammed parameter requiring dedicated selection mechanisms. An alternative view posits that movement duration emerges from the control policy itself. Here, we demonstrate, using infinite horizon optimal feedback control (IHOFC) and nonlinear limb dynamics, that this alternative hypothesis successfully captures diverse aspects of human reaching behavior, including trade-offs between movement duration and task parameters. Specifically, we reproduced the modulation of movement duration with varying reach distances and accuracy (Fitts's law) in the presence of nonlinear dynamics, and extended the infinite horizon framework to include the effect of rewards and biomechanical costs. Furthermore, our model also featured a temporal evolution of feedback responses to perturbations that resembles experimental observations and naturally accounted for motor decisions observed when participants select one among multiple goals in dynamic environments. Together, these developments show that in many cases, movement duration may not need to be specified a priori, but instead could result from task-dependent control policies. This framework validates a candidate explanation for varied movement durations, which invites a reconsideration of the nature and strength of evidence for the finite horizon formulation.

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23. Thousand-Brains Systems: Sensorimotor Intelligence for Rapid, Robust Learning and Inference

Niels Leadholm, Viviane Clay, Scott G Knudstrup, Hojae Lee, Jeff Hawkins
https://doi.org/10.1162/neco.a.1508 | pp. 845-896

Current AI systems achieve impressive performance on many tasks, yet they lack core attributes of biological intelligence, including rapid, continual learning, representations grounded in sensorimotor interactions, and structured knowledge that enables efficient generalization. Neuroscience theory suggests that mammals evolved flexible intelligence through the replication of a semi-independent, sensorimotor module, a functional unit known as a cortical column. To address the disparity between biological and artificial intelligence, Thousand Brains systems were proposed as a means of mirroring the architecture of cortical columns and their interactions. In our current work, we evaluate the unique properties of Monty, the first implementation of a Thousand Brains system. We focus on 3D object perception and, in particular, the combined task of object recognition and pose estimation. Utilizing the YCB data set of household objects, we first assess Monty's use of sensorimotor learning to build structured representations, finding that these enable robust generalization. These representations include an emphasis on classifying objects by their global shape, as well as a natural ability to detect object symmetries. We then explore Monty's use of model-free and model-based policies to enable rapid inference by supporting principled movements. We find that such policies complement Monty's modular architecture, a design that can accommodate communication between modules to further accelerate inference speed via a novel voting algorithm. Finally, we examine Monty's use of associative, Hebbian-like binding to enable rapid, continual, and computationally efficient learning, properties that compare favorably to current deep learning architectures. While Monty is still in a nascent stage of development, these findings support Thousand Brains systems as a powerful and promising new approach to AI and reinforce the importance of sensorimotor learning for developing intelligent systems.

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24. Implicit Generative Modeling by Kernel Similarity Matching

Shubham Choudhary, Paul Masset, Demba Ba

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Understanding how the brain encodes stimuli has been a fundamental problem in computational neuroscience. Insights into this problem have led to the design and development of artificial neural networks that learn representations by incorporating brain-like learning abilities. Recently, learning representations by capturing similarity among input samples has been studied (Pehlevan et al., 2018) to tackle this problem. This approach, however, has thus far been used only to learn downstream features from an input and has not been studied in the context of a generative paradigm, where one can map the representations back to the input space, incorporating not only bottom-up interactions (stimuli→latent) but also learning features in a top-down manner (latent→stimuli). We investigate a kernel similarity matching framework for generative modeling. Starting with a modified sparse coding objective for learning representations proposed in prior works (Olshausen & Field, 1996; Tolooshams & Ba, 2022), we demonstrate that representation learning in this context is equivalent to maximizing similarity between the input kernel and a latent kernel. We show that an implicit generative model arises from learning the kernel structure in the latent space and show how the framework can be adapted to learn manifold structures, potentially providing insights as to how task representations can be encoded in the brain. To solve the objective, we propose a novel alternate direction method of multipliers (ADMM)-based algorithm and discuss the interpretation of the optimization process. Finally, we discuss how this representation learning problem can lead toward a biologically plausible architecture to learn the model parameters that ties together representation learning using similarity matching (a bottom-up approach) with predictive coding (a top-down approach).

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25. Perceptual Processes as Charting Operators

Peter Neri

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Sensory operators are classically modeled using small circuits involving canonical computations, such as energy extraction and gain control. Notwithstanding their utility, circuit models do not provide a unified framework encompassing the variety of effects observed experimentally. We develop a novel, alternative framework that recasts sensory operators in the language of intrinsic geometry. We start from a plausible representation of perceptual processes that is akin to measuring distances over a sensory manifold. We show that this representation is sufficiently expressive to capture a wide range of empirical effects associated with elementary sensory computations. The resulting geometrical framework offers a new perspective on state-of-the-art empirical descriptors of sensory behavior, such as first-order and second-order perceptual kernels. For example, it relates these descriptors to notions of flatness and curvature in perceptual space.

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26. Comparing Dynamical Models Through Diffeomorphic Vector Field Alignment

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Dynamical systems models such as recurrent neural networks (RNNs) are increasingly popular in theoretical neuroscience as a vehicle for hypothesis generation and data analysis. Evaluating the dynamics in such models is key to understanding their learned generative mechanisms. However, such evaluation is impeded by two major challenges: (1) comparison of learned dynamics across models is difficult because a priori there is no enforced equivalence of their coordinate systems, and (2) identification of mechanistically important low-dimensional motifs (e.g., limit sets) is analytically intractable in high-dimensional nonlinear models such as RNNs. Here, we propose a comprehensive framework to address these two issues, termed diffeomorphic vector field alignment for learned models (DFORM). DFORM learns a nonlinear coordinate transformation between the state spaces of two dynamical systems, which aligns their trajectories in a maximally one-to-one manner. In so doing, DFORM enables an assessment of whether a set of models exhibits topological equivalence, that is, their dynamics are mechanistically similar despite differences in their coordinate systems. A by-product of this methodology is a means to locate dynamical motifs on low-dimensional manifolds embedded within higher-dimensional systems. We verified DFORM's ability to identify linear and nonlinear coordinate transformations using canonical topologically equivalent systems, RNNs, and systems related by nonlinear flows. DFORM was also shown to provide a quantification of similarity between topologically distinct systems. We then demonstrated that DFORM can locate important dynamical motifs including invariant manifolds and saddle limit sets within high-dimensional models. Finally, using a set of RNN models trained on human functional magnetic resonance imaging recordings, we illustrated that DFORM can identify limit cycles from high-dimensional data-driven models, which agreed well with prior numerical analysis.

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27. Toward Enhancing RMSProp With Forward-Looking Gradient Updates for Complex Loss Landscapes

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This letter introduces a novel algorithm for training deep neural networks with many nonlinear layers. Our method uses an approximated integrated gradient that is averaged over the range of the weight update to more accurately capture the loss change resulting from parameter updates. Unlike standard gradients, this average gradient improves learning efficiency in certain scenarios. We incorporate the approximated average gradient into RMSProp and compare the resulting algorithm to conventional RMSProp and Adam. We evaluate the approach on deep models lacking skip connections, such as those with many nonlinear activations and no residual structure, where traditional methods typically encounter difficulties. These models that focus on extracting high-order features create a loss landscape more akin to that of a biological brain. Our method requires significantly fewer iterations to reach a target training loss on MNIST, Fashion MNIST, and IMDb benchmarks for both convolutional and fully connected architectures across different initialization schemes. While our approach incurs moderately higher computational and memory costs compared to standard RMSProp, its performance on shallow models remains comparable. Nevertheless, our main contributions are (1) introducing the average gradient concept as an efficient alternative to computing high-order derivatives, (2) offering a novel factorization formula for approximating the average gradient, accompanied by a formal derivation, and (3) showing an example algorithm that leverages this formula to enhance the efficiency of RMSProp for some models, as validated by our evaluation.

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28. Graphon Signal Processing for Spiking and Biological Neural Networks

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Graph signal processing (GSP) extends classical signal processing to signals defined on graphs, enabling filtering, spectral analysis, and sampling of data generated by networks of various kinds. Graphon signal processing (GnSP) develops this framework further by employing the theory of graphons. Graphons are measurable functions on the unit square that represent graphs and limits of convergent graph sequences. The use of graphons provides stability of GSP methods to stochastic variability in network data and improves computational efficiency for very large networks. We use GnSP to address the stimulus identification problem (SIP) in computational and biological neural networks. The SIP is an inverse problem that aims to infer the unknown stimulus sfrom the observed network output f. We first validate the approach in spiking neural network simulations and then analyze calcium imaging recordings. Graphon-based spectral projections yield trial-invariant, low-dimensional embeddings that improve stimulus classification over principal component analysis and discrete GSP baselines. The embeddings remain stable under variations in network stochasticity, providing robustness to different network sizes and noise levels. To the best of our knowledge, this is the first application of GnSP to biological neural networks, opening new avenues for graphon-based analysis in neuroscience.

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29. Autonomous Learning With High-Dimensional Computing Architecture Similar to Von Neumann’s

Pentti Kanerva

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We model human and animal learning by computing with high-dimensional vectors (e.g., D = 10,000). The architecture resembles traditional (von Neumann) computing with numbers, but the instructions refer to vectors and operate on them in superposition. The architecture includes a high-capacity memory for vectors, counterpart of the random-access memory (RAM) for numbers. The model's ability to learn from data reminds us of deep learning, but with an architecture closer to biology. The architecture agrees with an idea from psychology that human memory and learning involve a short-term working memory and a long-term data store. Neuroscience provides us with a model of the long-term memory, namely, the cortex of the cerebellum. With roots in psychology, biology, and traditional computing, a theory of computing with vectors can help us understand how brains compute. Application to learning by robots seems inevitable, but there is likely to be more, including language. Ultimately we want to compute with no more material and energy than brains use. To that end, we need a mathematical theory that agrees with psychology and biology and is suitable for nano-technology. We also need to exercise the theory in large-scale experiments. The analogy with traditional computing suggests that the architecture be programmable in terms of variables, values, and data structures, the very things that have made traditional computing ubiquitous and that seem worth learning from and emulating.

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30. Sparse Graphical Modeling for Electrophysiological Phase-Based Connectivity Using Circular Statistics

Issey Sukeda, Takeru Matsuda

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Identifying phase coupling from electrophysiological signals recorded by multiple electrodes, such as electroencephalogram (EEG) and electrocorticography (ECoG), helps neuroscientists and clinicians understand underlying brain structures or mechanisms. From a statistical perspective, these signals are multidimensional circular measurements that are correlated with one another and can be effectively modeled using a torus graph model designed for circular random variables. Using the torus graph model avoids the issue of detecting spurious correlations. However, the naive estimation of this model tends to lead to a dense network structure that is difficult to interpret. Therefore, to enhance the interpretability of the brain network structure, this review proposes a sparse estimation method for the torus graph model using regularized score matching combined with information criteria. In numerical simulations, our method successfully recovered the true dependence structure from a synthetic data set. Furthermore, we present analyses of two real data sets, one involving human EEG and the other marmoset ECoG, demonstrating that our method can be widely applied to phase-coupling analysis across different types of neural data. We found that when we used our proposed method, the modularity of the estimated network structure revealed more resolved brain structures and demonstrated differences in trends among individuals.

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31. Global Stability of a Hebbian/Anti-Hebbian Network for Principal Subspace Learning

David Lipshutz, Robert J. Lipshutz

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Biological neural networks self-organize according to local synaptic modifications to produce stable computations. How modifications at the synaptic level give rise to such computations at the network level remains an open question. Pehlevan et al. (2015) proposed a model of a self-organizing neural network with Hebbian and anti-Hebbian synaptic updates that implements an algorithm for principal subspace analysis; however, global stability of the nonlinear synaptic dynamics has not been established. Here, for the case that the feedforward and recurrent weights evolve at the same timescale, we prove global stability of the continuum limit of the synaptic dynamics and show that the dynamics evolve in two phases. In the first phase, the synaptic weights converge to an invariant manifold where the neural filters are orthonormal. In the second phase, the synaptic dynamics follow the gradient flow of a nonconvex potential function whose minima correspond to neural filters that span the principal subspace of the input data.

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32. Domain Adaptation With Additional Features via Label-Aware and Graph-Based Fused Gromov-Wasserstein Optimal Transport

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In many domain adaptation tasks, the source and target domains share an identical feature space, so the domain gap arises only from the distributional shift. In practice; however, new-target-only features (e.g., sensors added after training) often become available at test time, violating the shared feature space assumption and invalidating most existing methods. We address this setting with Label-Aware and

Graph-Based Fused Gromov-Wasserstein Optimal Transport (LAGB-FGW), focusing on a transductive domain adaptation scenario, in which the entire unlabeled target data set is available during training, and predictions are jointly inferred for all target samples. LAGB-FGW (1) embeds label discrepancy directly into the source metric, (2) constructs a K-NN graph on the full target feature space to capture structure introduced by the additional features, and (3) jointly solves standard Optimal Transport (OT) and Gromov-Wasserstein OT, thereby transferring labels using both the common and the additional features. We validate LAGB-FGW on four synthetic benchmarks and the HAR70+ human-activity data set, and LAGB-FGW consistently outperforms all baselines, highlighting the advantage of combining source label information with graph-based structural cues when additional target features are available only at test time.

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33. Competition Between Memory Updating and Differentiation Emerges From Intrinsic Network Dynamics

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When an event occurs that is similar to a previous experience, the original episodic memory can be modified with new information (updating) or a new memory can be encoded separately (differentiation). Prediction errors, the deviation between expected and actual stimuli, are believed to mediate the competition between updating and differentiation, but the underlying mechanisms remain unclear. Here, we present a new analysis of experiments examining recognition memory and cued recall of similar conversations. We find that the original version is recognized more confidently than the modified version, the recognition confidence for modified versions shows a U-shaped dependence on the prediction error, and larger prediction errors lead to participants' more frequently recalling two versions. These results can be accounted for by a computational model based on a modified Hopfield network, which encodes the original and modified versions sequentially and weights the encoding of new patterns based on the prediction error. The model shows that (1) similar new memories interfere with previous ones (updating), while dissimilar ones are stored separately (differentiation); (2) interference from similar representations leads to reduced memory accuracy and lower-confidence recognition; and (3) modulation of the encoding strength is critical for accounting for the experimental data. Our modeling results show that prediction-error-driven competition between updating and differentiation can emerge from intrinsic network dynamics alone.

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34. Toward a Computational Phenomenology of Meditative Deconstruction: “Letting Go” and the Deconstruction of Experience With Active Inference

Shawn Prest

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Meditative experience has long been associated with conceptual attenuation, reduced reactivity to phenomena, increased present moment perception, and more pleasant experience. However, the computational mechanisms underlying such meditative deconstruction are not well understood, with no formal computational models available to explicate how deconstruction alters perception and action during meditation. Using the active inference framework, I demonstrate that the phenomenology of deconstruction-in terms of conceptual attenuation, reduced reactivity, and shorter temporal scale perception-naturally emerges from the dynamics of hierarchical inference when the deconstructive notion of

letting go is cast as a reduction in precision of beliefs about hidden states at a specific level of the generative model. I present a formal hierarchical three-level generative model and simulate deconstruction as an intervention in a facial recognition task, where the agent selects a letting-go policy when perceived affective valence becomes excessively negative. The results demonstrate that the capacity to deconstruct permits agents to self-regulate experienced affect via letting go. The model offers a novel perspective within the paradigm of computational phenomenology on conceptual attenuation, equanimity, stillness, and affect during meditative deconstruction.

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35. Cross-Frequency Coupling as a Neural Substrate for Prediction Error Evaluation: A Laminar Neural Mass Modeling Approach

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Abstract Predictive coding frameworks suggest neural computations rely on hierarchical error minimization, yet the neural implementation of this inference remains unclear. We propose that cross-frequency coupling (CFC) furnishes fundamental mechanism for this process. Using our laminar neural mass model (LaNMM), we demonstrate that signal-envelope coupling (SEC) an envelope-envelope coupling (EEC) instantiate a hierarchical comparator mechanism. Specifically, SEC generates prediction-error signals by subtracting top-down predictions from bottom-up oscillatory envelopes, while EEC operates at slower timescales to implement gating—a critical mechanism for precision weighting. To establish the face validity and clinical implications of this proposal, we model perturbations of these CFC mechanisms to investigate their roles in pathophysiological and altered neuronal function. In Alzheimer's disease, interneuron dysfunction disrupts the comparator, causing aberrant prediction error amplification in early stages and drastic attenuation in late phases. Conversely, serotonergic psychedelics increase excitatory gain, diminishing the modulatory effect of predictions and causing a failure of sensory attenuation. Collectively, these findings implicate multiscale cross-frequency coupling as a key computational mechanism supporting predictive coding and its disruption in disease.

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36. A Model-Free Reinforcement Learning Implementation of Decision Making Under Uncertainty by Sequential Sampling

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Although evidence integration to the boundary model has successfully explained a wide range of behavioral and neural data in decision making under uncertainty, how animals learn and optimize the boundary remains unresolved. Here, we propose a model-free reinforcement learning algorithm for perceptual decisions under uncertainty that implements a sequential sampling process with an implicit decision boundary. Our model learns whether to commit to a decision given the available evidence or continue sampling information at a cost. We reproduced the canonical features of perceptual decision making such as dependence of accuracy and reaction time on evidence strength, modulation of speed-accuracy trade-off by payoff regime, and many others. By unifying learning and decision making within the same framework, this model can inspire a new look at the mechanisms of flexibility in context-dependent changes of behavior.

37. DROP: Distributional and Regular Optimism and Pessimism for Reinforcement Learning

Taisuke Kobayashi

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In reinforcement learning (RL), temporal difference (TD) error is known to be related to the firing rate of dopamine neurons. It has been observed that each dopamine neuron does not behave uniformly, but each responds to the TD error in an optimistic or pessimistic manner, interpreted as a kind of distributional RL. To explain such biological data, a heuristic model has also been introduced with learning rates asymmetric for the positive and negative TD errors. However, this heuristic model is not theoretically grounded, and it is unknown whether it can work as an RL algorithm. This letter therefore introduces a novel theoretically grounded model with optimism and pessimism, which is derived from control as inference. In combination with ensemble learning, a distributional value function as a critic is estimated from regularly introduced optimism and pessimism. Based on its central value, a policy in an actor is improved. This proposed algorithm, so-called DROP (distributional and regular optimism and pessimism), is compared on dynamic tasks. Although the heuristic model showed poor learning performance, DROP demonstrated excellent performance in all tasks with high generality. In addition, DROP achieved learning performance comparable to the state-of-the-art algorithms. In other words, it was suggested that DROP is a new model that can elicit the potential contributions of optimism and pessimism.

38. W-Kernel and Its Principal Space for Frequentist Evaluation of Bayesian Estimators

Yukito Iba

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Evaluating the variability of posterior estimates is a key aspect of Bayesian model assessment. In this study, we focus on the posterior covariance matrix W, defined through the log likelihoods of individual observations. Previous studies, notably MacEachern and Peruggia (2002) and Thomas et al. (2018), examined the role of the principal space of W in Bayesian sensitivity analysis. Here, we show that the principal space of W is also central to frequentist evaluation, using the recently proposed Bayesian infinitesimal jackknife (Bayesian IJ) approximation (Giordano & Broderick, 2023) as a key tool. We further clarify the relationship between W and the Fisher kernel, showing that a modified version of the Fisher kernel can be viewed as an approximation to W. Moreover, the matrix W itself can be interpreted as a reproducing kernel, which we refer to as the W-kernel. Based on this connection, we investigate the relation between the W-kernel formulation in the data space and the classical asymptotic formulation in the parameter space. We also introduce the matrix Z, which is effectively dual to W in the sense of PCA; this formulation provides another perspective on the relationship between W and classical asymptotic theory. In the appendixes, we explore approximate bootstrap methods for posterior means and show that projection onto the principal space of W facilitates frequentist evaluation when higher-order terms are included. In addition, we introduce incomplete Cholesky decomposition as an efficient method for computing the principal space of W and discuss the concept of representative subsets of observations.

39. A Hidden Markov Model–Inspired Sequence Classification Method for Hyperdimensional Computing

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This letter introduces a novel method for discrete-sequence classification within the hyperdimensional computing (HDC) paradigm. The method, inspired by the concept of hidden Markov models (HMM), implements mechanisms that effectively address the major challenges arising in analyzing sequences generated by real-world processes, such as variable length, misalignment, symbol repetitions, omissions, or insertions. All algebraic transformations of input data in HMM-based sequence classification are replaced by bitwise operations on appropriately derived hyperdimensional binary vectors (hypervectors). The proposed hypervector transformation pipeline mirrors the algebraic manipulations of the HMM paradigm and involves a procedure that prevents information decay when processing longer sequences. Since only binary bit-wise operations on hypervectors are used in computations, the proposed method is well suited for hardware implementation, especially in the form of custom in-memory computing VLSI devices. The experimental evaluation of the proposed method, conducted on both artificial and real-world data sets, demonstrates its superiority over existing HDC-based sequence analysis approaches, yielding noticeable gains in classification accuracy. We also demonstrate that the proposed approach, like most HDC-based sequence classification methods, is robust against adverse physical, internal, or external factors that induce bit flips in the underlying hardware, which could entirely corrupt the outcomes of classical sequence-analysis methods, such as HMMs.